



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME:

Sentinel



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The problem statement

Overcrowded emergency departments depend on one nurse to rank every patient.

When an emergency department is overwhelmed, the order in which patients are seen depends on a single triage nurse working under pressure, with seconds and incomplete information per patient. Most of the time this works. The dangerous exception is under-triage: a genuinely sick patient who looks stable is marked non-urgent, left waiting, and gets worse before anyone notices. Under-triage is far more harmful than over-triage, and it is the easiest mistake to make on a crowded floor.

Under-triage

Marking a sick patient non-urgent is the deadliest triage error, and the easiest to make when the floor is full and every second counts.

No re-checks

Most triage happens once, at arrival, then stops. A patient who is stable at check-in can worsen in the waiting room with no one watching.

Impact: longer waits, patients who leave without being seen, and preventable harm.

Who it hurts: every arriving patient, and the nurses held responsible for the call.

Our Proposed solution:

Sentinel is three AI agents in a chain, and the hospital decides how much they do on their own.

AGENT 1: INTERPRETER

GETS raw, often sparse intake: complaint, appearance, arrival mode, vitals (BP, SpO2, HR, temp), meds, allergies, history, pre-hospital care, times.

DOES structures the data; computes shock index (HR over systolic BP) and time since onset; maps vitals to danger zones; scores what is missing.

OUTPUTS a clean snapshot: findings, derived indices, a red flag list, the data gaps, and a confidence score.

AGENT 2: ADJUDICATOR

GETS the snapshot, once the nurse has approved it.

DOES assigns an ESI acuity level; runs the clock against time critical windows (stroke 4.5 h, STEMI 90 min, sepsis 1 h); sets a monitoring tier.

OUTPUTS acuity 1 to 5, placement, a monitoring tier, the top 3 drivers, a what-if-ignored note, and a confidence score.

AGENT 3: WATCHER

GETS the confirmed acuity and monitoring tier for everyone now waiting.

DOES spends a limited attention budget by risk; re-checks the sick often, the stable rarely (15 min to 2 h by acuity); takes patient self-report; a timer backstop catches anyone past a safe wait.

OUTPUTS live re-ranking: pushes a patient up and alerts the nurse, with the reason, the moment they worsen or wait too long. Never down.

CONTINUOUS LEARNING LOOP (runs every shift; the system keeps getting better and earns more autonomy over time):

1. LOG every decision, every nurse override, and the patient's real outcome.



2. LEARN retrain and re-calibrate confidence, federated across hospitals, with no patient data moved.



3. IMPROVE sharper acuity calls and fewer corrections, which unlocks the next autonomy level.

AUTONOMY THE HOSPITAL CONTROLS (dial it up as the loop proves the system):

L1 ASSIST

AI suggests. Nurse decides and confirms every step.

L2 PREPARE

AI does safe prep and standing orders. Nurse confirms each patient.

L3 AUTO-ADVANCE

AI clears clear-cut cases itself. Nurse reviews the exceptions.

L4 SELF-RUN

AI runs monitoring and surge. Nurse oversees, can override.

WHO IT HELPS

The patient who looked fine and would have been missed.
The nurse who no longer carries the room alone. The family who gets to take their person home.

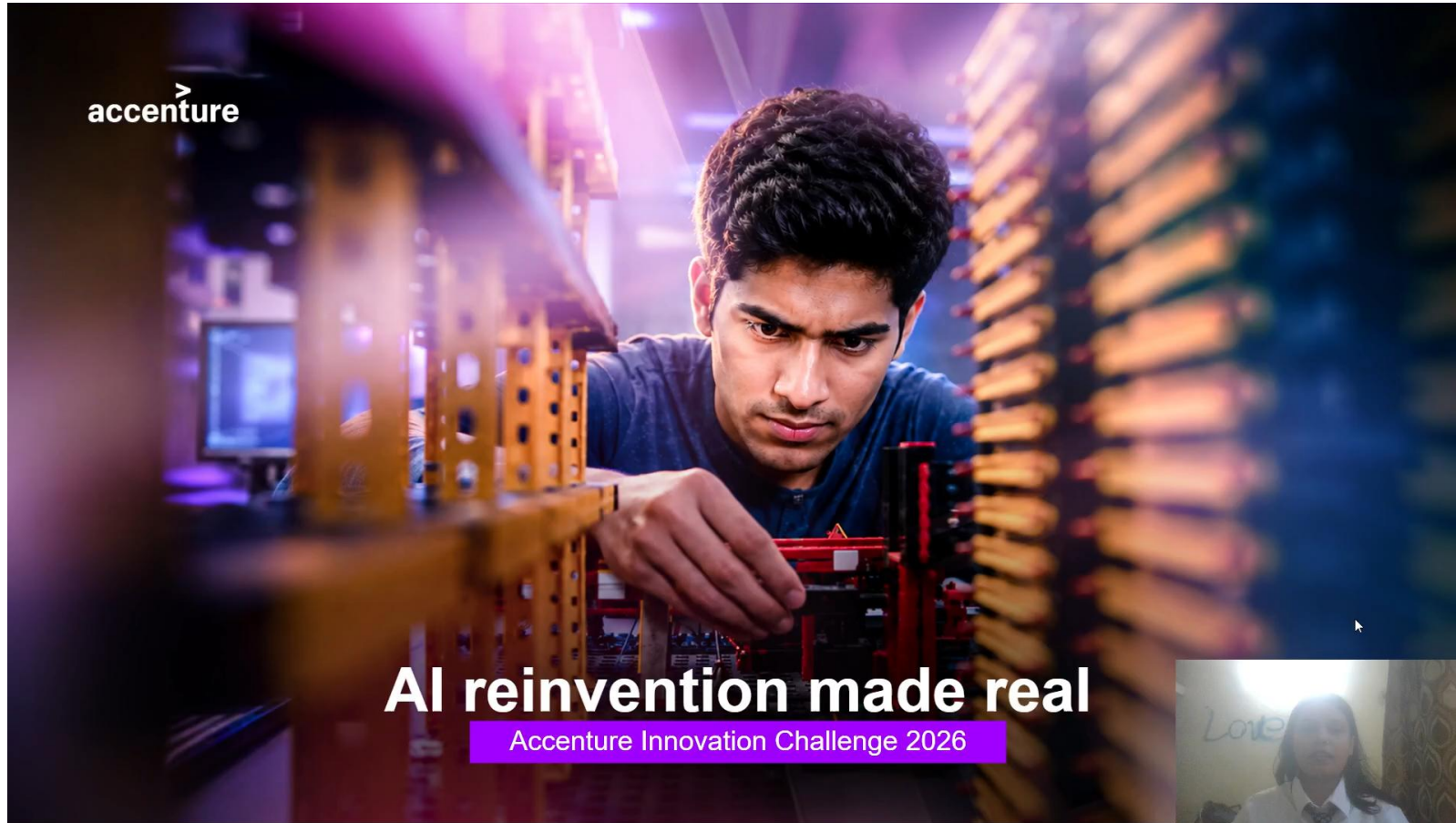
WHY IT MATTERS

It saves lives. Faster, fairer triage reaches the sickest first and cuts human error, so no one fades unnoticed. Minutes are everything.

BUSINESS IMPACT

Shorter waits, fewer patients who leave without care, fewer costly misses. A safety net that never tires and scales to every hospital.

Video Walkthrough



This video walkthrough the problem, the Priority Ratchet, and the three agent pipeline in action. The full talk track is provided alongside this deck.

Thank you

